

FUNCTIONAL REQUIREMENTS DOCUMENT

RECRUIT-IQ

AI-Augmented Recruitment and Interview Platform

Item	Details
Document owner	Suvetha S P
Version	1.0 (Draft for review)
Date	April 2026
Related document	RECRUIT-IQ Business Requirements Document v1.0 (requirement IDs BR-xx and FR-xx are used in both)
Live system	https://recruit-iq-ten.vercel.app
Source code	https://github.com/Suvetha29/RECRUIT-IQ

1. Introduction

1.1 Purpose

This document turns the business requirements in the BRD into detailed functional specifications, what each function does, who can use it, what goes in, what comes out and what happens when something goes wrong. Developers use it to build, testers use it to write test cases and stakeholders use it to confirm the behaviour.

1.2 Scope

It covers the release-1 features of RECRUIT-IQ: authentication and roles, job management, applications and ATS scoring, the hiring pipeline, notifications and email, assessments, interview evaluation and offer letters.

1.3 Intended audience

- Developers and testers building and verifying the system
- Product owner and HR stakeholders confirming the behaviour
- Reviewers assessing the completeness of the requirements

1.4 References

- RECRUIT-IQ Business Requirements Document v1.0
- Project README and API documentation (UI at /docs when the backend is running)

2. System Overview

RECRUIT-IQ is a web application. A React single-page frontend talks to a FastAPI REST backend, which stores data in a relational database and connects to external services for AI, email and video meetings.

Component	Responsibility
Web frontend (React)	Screens for HR and candidates; sends requests with the user's access token
Backend API (FastAPI)	Business logic, role checks, ATS scoring, assessments, evaluation pipeline, PDF generation
Database (SQLAlchemy)	Users, jobs, applications, assessments, results, notifications
File storage	Resumes, interview recordings and offer letters
AI service (LLM and speech-to-text)	Job content generation, interview transcription and evaluation
Email server (SMTP)	Status emails and offer letters
Video meeting service (Jitsi Meet)	Interview rooms

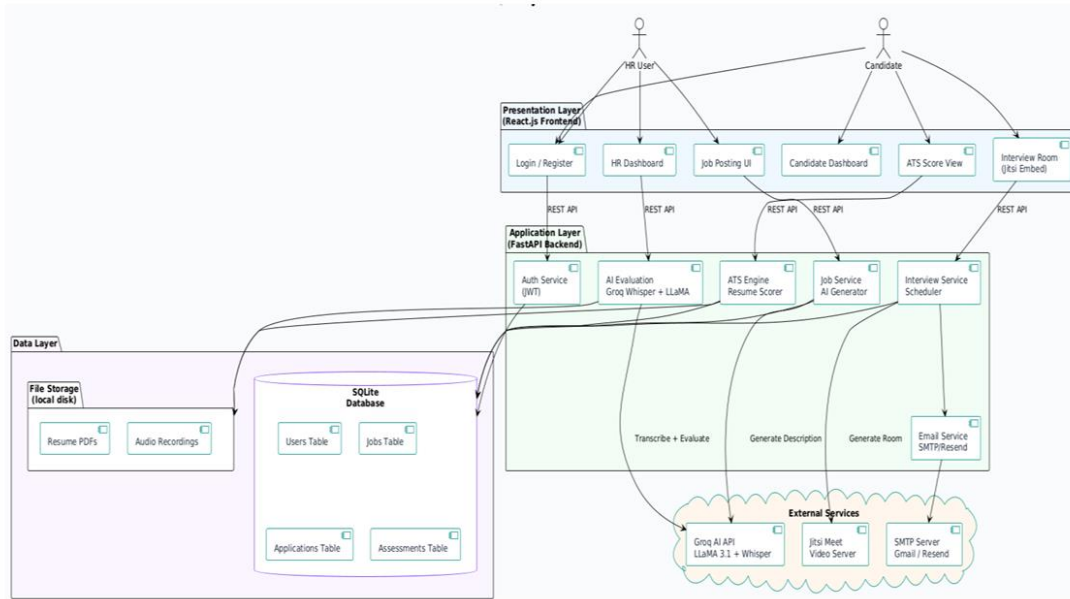


Figure 2.1 system architecture

3. Users and Permissions

Function	Visitor	Candidate	HR
Register and log in	Yes	Yes	Yes
Create, edit, close, delete jobs	No	No	Yes
Generate job content with AI	No	No	Yes
Browse open jobs and apply	No	Yes	No
View own applications	No	Yes	No
View applicants and ATS scores	No	No	Yes
Change application status	No	No	Yes
Create assessments	No	No	Yes
Take assessments	No	Yes	No
View detailed assessment results	No	No	Yes
Upload interview audio and view evaluation	No	No	Yes
Generate and send offer letters	No	No	Yes
View own notifications	No	Yes	Yes

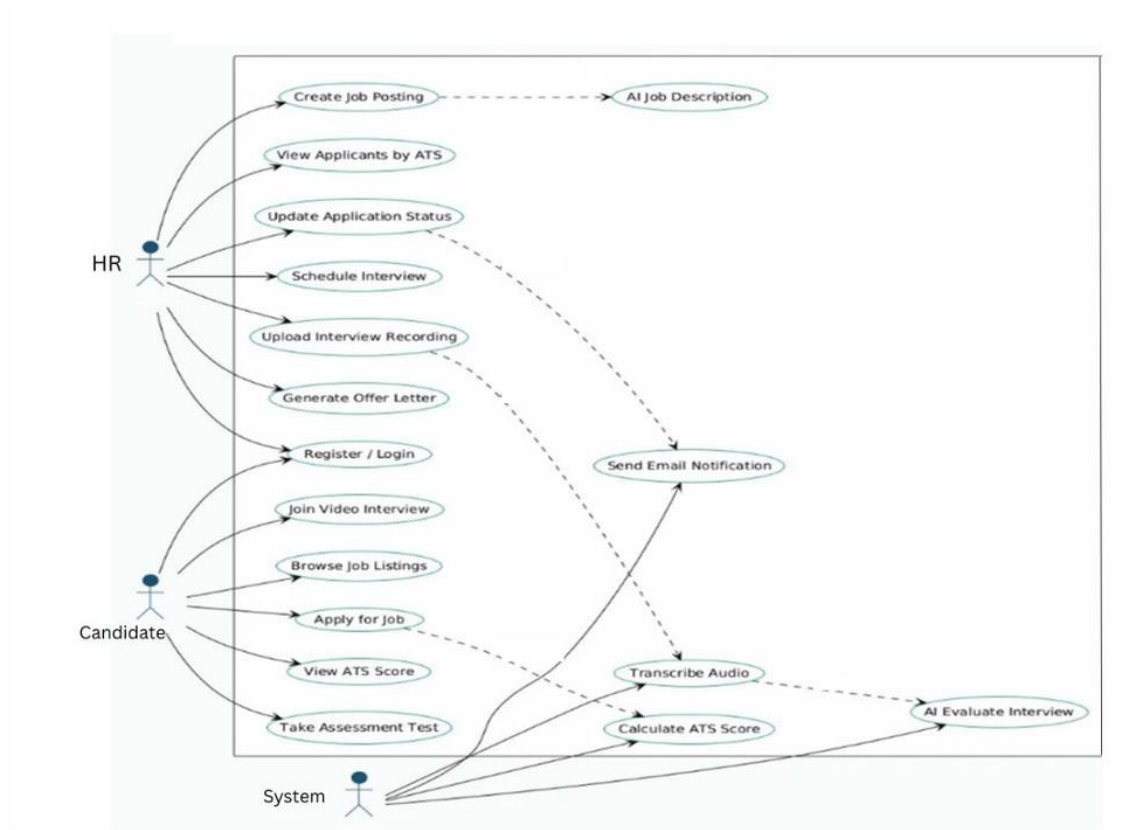


Figure 3.1 Use Case Model

4. Functional Specifications

4.1 Authentication and access control

FR-01 User registration and login	
Actor	Visitor (not logged in)
Preconditions	None
Main flow	1. The user opens the Register or Login screen. 2. The user enters email and password. On registration the user also chooses a role. 3. The system validates the input and creates the account (register) or checks the password (login). 4. The system returns a signed access token. 5. The app keeps the token and sends it with every later request.
Rules	Email must be in a valid format. Passwords are stored hashed, never as plain text. Tokens are signed and stateless.
Outputs	Access token; user is taken to the screens for their role
Exceptions	Invalid email format: validation message. Wrong email or password: login rejected. Invalid or expired token on later requests: 401 Unauthorized, user must log in again.
Interfaces	POST /api/auth/register, POST /api/auth/login

FR-02 Role-based access control	
Actor	Any logged-in user
Preconditions	Valid access token
Main flow	1. Each request carries the access token. 2. The server identifies the user and their role. 3. The server compares the role with the role the function requires. 4. The request is allowed or denied.
Rules	HR-only and candidate-only functions are listed in section 3. Checks are made on the server for every protected function.
Outputs	Requested data or action, or a denial
Exceptions	Missing or invalid token: 401. Wrong role: 403 Forbidden (for example, a candidate calling an HR function).

4.2 Job management

FR-03 Create and manage job postings	
Actor	HR
Preconditions	Logged in as HR
Main flow	1. HR opens the job form and enters title, description, requirements and responsibilities. 2. HR saves. The job is created as open and linked to the HR user who created it. 3. HR can later edit the job, close it or delete it.
Rules	A job is either open or closed. Only open jobs can receive applications. Deleting a job also deletes its assessment, assessment results and applications.
Outputs	Saved job and updated job lists
Exceptions	Non-HR user: 403. Missing required fields: validation message.
Interfaces	Job endpoints as listed in the API documentation (/docs)

FR-04 Generate job content with AI	
Actor	HR
Preconditions	Logged in as HR; AI service reachable
Main flow	1. HR enters basic job details, such as the title and key points. 2. HR chooses Generate. 3. The system sends a structured request to the AI service asking for JSON output. 4. The AI returns a description, requirements and responsibilities. 5. The fields are filled in on the form. HR edits them and saves.
Rules	The output must be structured JSON with three parts. Nothing is saved until HR saves the job. HR can always type the content manually.
Outputs	Draft description, requirements and responsibilities
Exceptions	AI service unavailable or rate limited: a clear message is shown and HR continues manually.
Interfaces	POST /api/jobs/generate

4.3 Applications and the hiring pipeline

FR-05 Browse jobs and apply	
Actor	Candidate
Preconditions	Logged in as candidate; job is open
Main flow	1. The candidate opens the list of open jobs and selects one. 2. On the apply screen the candidate uploads a PDF resume. 3. The candidate submits. 4. The system stores the file, extracts the text, calculates the ATS score (FR-06) and creates the application with status Pending. 5. A confirmation is shown and the application appears under My Applications.
Rules	The resume must be a PDF. The application links the candidate, the job, the resume file and the score.
Outputs	New application with status Pending and an ATS score
Exceptions	Non-candidate user: 403. Missing or non-PDF file: validation message. Resume with no readable text: the application is saved with a low score.
Interfaces	POST /api/applications/apply (multipart form)

FR-06 Calculate the ATS score	
Actor	System (runs when an application is submitted)
Inputs	Resume text and the job's description, requirements and responsibilities
Main flow	1. Extract text from the resume PDF. 2. Keyword match: split resume and job text into words of three or more letters, remove common stop words and measure the overlap. 3. Skill match: look for terms from a fixed list of about 40 common skills in both resume and job. 4. Experience match: find patterns such as "3 years" or "5+ years" in resume and job and compare them, capped at 100%. 5. Combine the three results with the fixed weights. 6. Save the score on the application.
Rules	$\text{Score} = 0.50 \times \text{skill match} + 0.30 \times \text{keyword match} + 0.20 \times \text{experience match}$, each as a percentage.
Outputs	ATS score stored with the application
Limits	Rule-based, so it depends on exact keywords. Synonyms and implied skills are not recognised. The score orders applicants; it never rejects anyone automatically.

FR-07 View ranked applicants	
Actor	HR
Preconditions	Logged in as HR
Main flow	1. HR opens a job and selects its applicants. 2. The system lists applicants ordered by ATS score, highest first. 3. HR can also open the combined list of all applicants.
Outputs	Applicant list with score and current status
Exceptions	Non-HR user: 403.

Interfaces	GET /api/applications/job/{job_id}
-------------------	------------------------------------

FR-08 Update application status	
Actor	HR
Preconditions	Application exists
Main flow	1. HR opens an application in the list or on the Kanban board. 2. HR selects the new status. 3. The system saves it. 4. The system creates a notification for the candidate (FR-10). 5. The system sends a status email in the background (FR-11).
Rules	Allowed statuses: Pending, Under review, Shortlisted, Interview, Hired, Rejected. The normal order is Pending, Under review, Shortlisted, Interview, then Hired or Rejected.
Outputs	Updated status, notification and email
Exceptions	Non-HR user: 403. Status outside the allowed list: rejected.
Interfaces	PATCH /api/applications/{id}/status

FR-09 View my applications	
Actor	Candidate
Preconditions	Logged in as candidate
Main flow	1. The candidate opens My Applications. 2. The system lists only that candidate's applications with job and current status.
Outputs	List of own applications
Exceptions	HR token: 403 (the function is for candidates only).
Interfaces	GET /api/applications/my

4.4 Notifications and email

FR-10 In-app notifications	
Actor	System; HR and candidates read them
Trigger	Events such as a status change
Main flow	1. The system creates a notification for the affected user. 2. A notification stores the message and, where relevant, the related job and application. 3. The notification bell on the page checks for new notifications every 30 seconds. 4. Selecting a notification opens the related screen for that user's role.
Outputs	Unread notifications shown on the bell
Rules	Notifications are shown only to the user they belong to.

FR-11 Status emails	
Actor	System
Trigger	Application status change

Main flow	1. After the status is saved, the system prepares an email to the candidate. 2. The email is sent in a background task so HR is not kept waiting.
Rules	Emails need an email account configured for the system.
Exceptions	If email settings are missing, a warning is logged, no email is sent and the status change still succeeds.

4.5 Assessments

FR-12 Create an assessment	
Actor	HR
Preconditions	Job exists
Main flow	1. HR opens the assessment screen for a job. 2. HR adds multiple-choice questions, each with answer options and the correct option. 3. HR sets the passing score and saves.
Rules	One assessment per job. Saving again updates the existing one. Questions are stored as structured data with the correct option recorded.
Outputs	Saved assessment
Exceptions	Non-HR user: 403.
Interfaces	POST /api/assessments/create

FR-13 Take an assessment	
Actor	Candidate
Preconditions	Job has an assessment; the candidate has an application
Main flow	1. The candidate opens the timed assessment screen. 2. The candidate selects one option per question and submits. 3. The server compares the answers with the stored correct options and calculates the score. 4. The server compares the score with the passing score and records pass or fail. 5. The candidate sees the result.
Rules	Scoring is done only on the server. The result is stored once per application.
Outputs	Score and pass or fail
Exceptions	Non-candidate user: 403.
Interfaces	POST /api/assessments/submit

FR-14 View detailed assessment results	
Actor	HR
Preconditions	Candidate has submitted the assessment
Main flow	1. HR opens a candidate's assessment result. 2. The system shows each question with the candidate's answer and the correct answer.
Outputs	Question-by-question breakdown
Exceptions	Non-HR user: 403.
Interfaces	GET /api/assessments/results/{application_id}/detailed

4.6 Interviews and evaluation

FR-15 Interview video link	
Actor	HR
Preconditions	Application is at the interview stage
Main flow	1. HR requests an interview link for the application. 2. The system generates a meeting room address for that application. 3. The link is shown to HR and made available to the candidate.
Rules	The room name is generated in a repeatable way so the same application always gets the same room.
Outputs	Video meeting link

FR-16 Interview audio evaluation	
Actor	HR
Preconditions	Interview recording available
Main flow	1. HR uploads the interview audio for an application. 2. The system stores the recording. 3. The speech-to-text service turns it into a transcript. 4. The AI service evaluates the transcript and returns a structured result: hire or reject recommendation, strengths, weaknesses and a score. 5. The transcript and AI score are saved on the application and shown to HR.
Rules	The evaluation supports the recruiter. The final decision is always made by HR.
Outputs	Transcript, AI score, recommendation, strengths and weaknesses
Exceptions	Audio cannot be read or an AI service error: an error message is shown and nothing is overwritten.
Interfaces	POST /api/evaluation/upload

4.7 Offer letters

FR-17 Generate and send an offer letter	
Actor	HR
Preconditions	Application exists
Main flow	1. HR chooses Generate offer letter for an application. 2. The system builds a PDF letter filled with the candidate and job details. 3. The PDF is saved and returned to HR to download. 4. The system emails the PDF to the candidate as an attachment in the background.
Rules	The letter uses one standard template for all candidates.
Outputs	Offer letter PDF and email
Exceptions	If email settings are missing, the PDF is still generated and a warning is logged.
Interfaces	POST /api/evaluation/offer-letter/{application_id}



Figure 4.8 Activity diagram

5. Screens

Screen	User	Purpose	Requirements
Register / Login	Visitor	Create an account and sign in	FR-01
Job form and job list	HR	Create and manage jobs, generate content with AI	FR-03, FR-04
Apply for a job	Candidate	Upload resume and submit	FR-05
My Applications	Candidate	See status of own applications	FR-09
Applications for a job	HR	Ranked applicants and status changes	FR-06, FR-07, FR-08
All Applicants	HR	Combined view of all applicants	FR-07
Kanban board	HR	Move applications across pipeline stages	FR-08
Assessment setup	HR	Create the multiple-choice test	FR-12, FR-14
Assessment (timed)	Candidate	Answer the test	FR-13
Interview evaluation	HR	Upload audio, view transcript and evaluation	FR-15, FR-16

Screen	User	Purpose	Requirements
Notification bell	HR, Candidate	See updates and open the related screen	FR-10

6. Data Specification

Table	Key fields	Relationships
users	email, hashed password, role (HR or candidate)	One HR user can own many jobs; one candidate can have many applications
jobs	title, description, requirements, responsibilities, status (open or closed), recruiter	One job has many applications and one assessment
applications	status, ATS score, AI score, recording path, transcript, resume file	Links one candidate to one job
assessments	questions (structured list), passing score	One per job
assessment_results	answers (list of chosen options), score, passed	One per application
notifications	message, recipient, job and application references (optional)	Used for links to the related screen

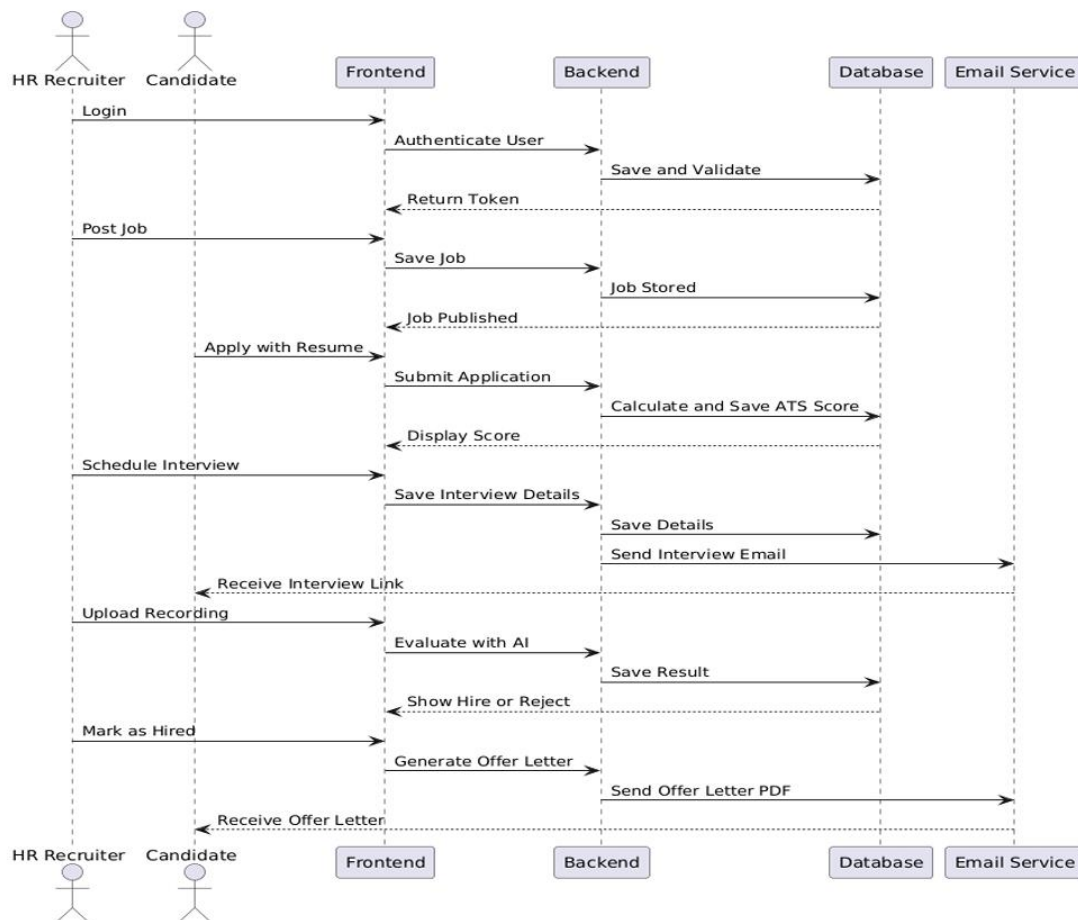


Figure 3.3.2 Sequence diagram

7. Error Handling

Situation	System behaviour
Missing or invalid access token	Request refused with 401; the user is asked to log in again
Function not allowed for the user's role	Request refused with 403 Forbidden
Invalid input (wrong format, missing field)	Request refused with a validation message that names the problem
AI service unavailable or rate limited	Clear message to the user; manual alternative remains available
Email settings missing or sending fails	Warning is logged; the main action still completes
Resume contains no readable text	Application saved; score is low; HR can still open the resume
Record not found	Not found message; no data changed

8. Test Scenarios

ID	Scenario	Expected result	Req.
TC-01	Register with a valid email and password	Account created and token returned	FR-01
TC-02	Log in with a wrong password	Login rejected	FR-01
TC-03	Candidate calls an HR function	403 Forbidden	FR-02
TC-04	HR generates job content	Description, requirements and responsibilities are filled in and editable	FR-04
TC-05	Candidate applies with a PDF resume	Application created as Pending with an ATS score	FR-05, FR-06
TC-06	HR opens applicants for a job	Ordered by ATS score, highest first	FR-07
TC-07	HR changes a status	Status saved, notification created, email sent	FR-08, FR-10, FR-11
TC-08	Candidate opens My Applications	Only own applications with current status	FR-09
TC-09	HR saves an assessment twice for one job	One assessment exists, updated by the second save	FR-12
TC-10	Candidate submits answers	Score calculated on the server; pass or fail shown	FR-13
TC-11	HR opens detailed results	Candidate answer and correct answer shown per question	FR-14
TC-12	HR uploads interview audio	Transcript and structured evaluation saved	FR-16
TC-13	HR generates an offer letter	PDF created and emailed as an attachment	FR-17
TC-14	Change a status with no email settings	Status saved; warning logged; no failure	FR-11
TC-15	HR deletes a job	Its assessment, results and applications are removed	FR-03

9. Approval

Name	Role	Signature	Date
[Name]	Business analyst and project owner	[Signature]	[Date]
[Name]	Reviewer	[Signature]	[Date]